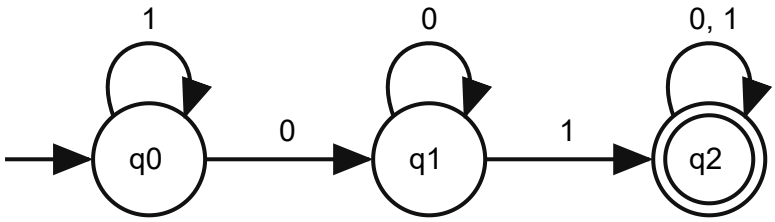
FLAT – Unit 1 Quiz (SemPrep sample)

This box and the title are ignored by SemPrep, because only numbered questions are read.
Each question: the question line, then a diagram if needed, then options A to D, then Answer. Topic and Explanation are optional.
Put each diagram as a picture between the question and option A. Save the document as PDF and upload it in Admin, Quiz bank.

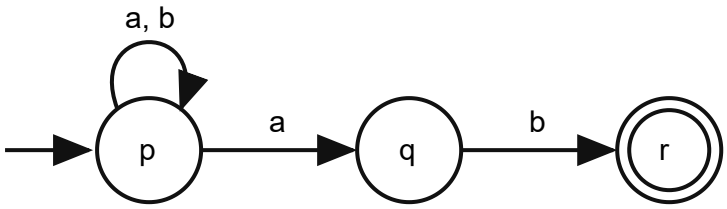
1. Which language over {0, 1} is accepted by the DFA shown below?



A) Strings that contain the substring 01
B) Strings that end with 10
C) Strings with an even number of 1s
D) Strings that start with 0
Answer: A
Topic: DFA
Explanation: q2 is reached only after a 0 is followed by a 1, and q2 loops on every symbol, so any string containing 01 is accepted.

2. How many start states does a DFA have?
A) Exactly one
B) At least two
C) Zero or more
D) One for each input symbol
Answer: A
Topic: DFA
Explanation: A DFA is a 5-tuple with a single start state q0.

3. For the NFA below, which state set is reached from {p} on input "ab"?



A) {p, r}
B) {q}
C) {p, q, r}
D) {r}
Answer: A
Topic: NFA
Explanation: On a: {p, q}. On b from {p, q}: p goes to p and q goes to r, giving {p, r}.

4. In the transition table below (start state A, final state C), what is δ(A, 1)?

State	0	1
→ A	B	A
B	B	C
* C	C	C

A) A
B) B
C) C
D) Undefined
Answer: A
Topic: Transition table

5. Which of these is NOT a regular language?
A) All strings over {a, b} ending in b
B) { a^n b^n | n ≥ 0 }
C) Strings with an even number of a's
D) The empty language
Answer: B
Topic: Regular languages
Explanation: Matching equal counts of a and b needs unbounded memory; the pumping lemma shows it is not regular.